

Weather Data Analytics Dashboard (Power BI Project)

Project Overview

This project focuses on analysing multi-dimensional weather data to uncover meaningful environmental patterns across districts and mandals. Using **Power BI**, this solution transforms raw weather observations into interactive dashboards that help monitor rainfall trends, temperature behaviour, humidity fluctuations, wind speed patterns, and overall weather performance.

The project is designed to simulate **real-world climate monitoring and environmental decision-making systems**, helping stakeholders identify seasonal patterns, extreme weather events, and district-level weather performance.

Dataset Overview

The dataset contains structured weather observation data collected across multiple geographical regions.

Dataset Description

- Total Records: **6.67 Lakhs+ weather observations**
- Time Period Covered: **Multiple Years (Daily Observations)**
- Geographic Coverage:
 - District Level Monitoring
 - Mandal Level Monitoring

Data Fields

Column	Description
District	Regional administrative division
Mandal	Sub-region within districts
Date	Weather observation date
Rain (mm)	Rainfall measured in millimetres
Min Temp (°C)	Minimum recorded temperature
Max Temp (°C)	Maximum recorded temperature
Min Humidity (%)	Minimum humidity level
Max Humidity (%)	Maximum humidity level
Min Wind Speed (Kmph)	Minimum wind speed
Max Wind Speed (Kmph)	Maximum wind speed

Data Cleaning & Preparation

To ensure reliable analytics, multiple data transformation steps were performed using **Power Query Editor**.

Data Cleaning Steps Performed

- Standardized data types for numerical and date columns.
- Removed duplicate weather records.
- Filtered null and missing environmental values.
- Trimmed and cleaned district and mandal text fields.
- Validated logical data consistency (Min values never exceed Max values).
- Created seasonal classification based on observation month.
- Generated derived metrics such as:
 - Temperature Range
 - Humidity Range
 - Wind Speed Range
 - Time Intelligence fields (Year, Month, Season)

Dashboard 1 – Rainfall Analysis

KPI Cards

Total Rainfall

Displays cumulative rainfall across selected regions and time periods.

Helps monitor overall precipitation intensity.

Average Rainfall

Shows the average rainfall per observation period, helping understand normal rainfall levels.

Maximum Rainfall Recorded

Identifies peak rainfall events and helps detect extreme weather occurrences.

Annual Rainfall

Tracks yearly rainfall accumulation to analyse climate changes over time.

Rainfall Growth Percentage

Measures rainfall increases or decreases compared to previous years.

Visual Insights

1 Yearly Rainfall Trend Comparison Across Districts

Shows long-term rainfall behaviour and highlights climate shifts.

2 Monthly Rainfall Distribution by Mandal

Helps identify local rainfall patterns and regional seasonal dependencies.

3 Seasonal Rainfall Variation by District

Visualizes monsoon and post-monsoon rainfall dominance.

4 Rainfall Intensity Monitoring

Identifies sudden rainfall spikes and anomaly events.

5 Rainfall Percentage Contribution by Month

Analyses how annual rainfall distributes across months.

Dashboard 2 – Temperature Insights

KPI Cards

Average Minimum Temperature

Tracks baseline cooling trends across districts.

Average Maximum Temperature

Measures peak heat levels and summer intensity.

Average Temperature Range

Highlights daily temperature fluctuations and stability.

Highest Temperature Recorded

Detects extreme heat events.

Lowest Temperature Recorded

Identifies cold wave patterns.

Visual Insights

1 Yearly Temperature Trend Comparison

Reveals warming or cooling climate shifts across districts.

2 Seasonal Maximum Temperature Variation

Highlights temperature extremes across seasons.

3 Monthly Temperature Heatmap

Identifies peak heat months geographically.

4 Temperature Range Distribution

Shows daily temperature volatility.

5 Extreme Temperature Pattern Detection

Helps identify abnormal climate days and district variations.

Dashboard 3 – Humidity Patterns

KPI Cards

Average Minimum Humidity

Tracks driest atmospheric conditions.

Average Maximum Humidity

Measures highest moisture concentration levels.

Humidity Range

Highlights daily moisture variation.

Highest Humidity Recorded

Detects extreme humidity events.

Lowest Humidity Recorded

Identifies dry atmospheric periods.

Visual Insights

1 Yearly Humidity Trend Analysis

Shows long-term atmospheric moisture patterns.

2 Seasonal Humidity Distribution

Reveals humidity peaks during monsoon seasons.

3 Rainfall vs Humidity Correlation

Shows relationship between precipitation and moisture.

4 Monthly Humidity Heatmap

Displays humidity concentration throughout the year.

5 Humidity Stability Analysis

Measures environmental consistency.

Dashboard 4 – Wind Speed Analysis

KPI Cards

Average Minimum Wind Speed

Measures calm wind conditions.

Average Maximum Wind Speed

Tracks peak wind intensity across regions.

Wind Speed Range

Analyses wind fluctuation behaviour.

Highest Wind Speed Recorded

Identifies extreme wind or storm events.

Lowest Wind Speed Recorded

Highlights still weather conditions.

Visual Insights

1 Yearly Wind Speed Trend

Reveals long-term wind pattern changes.

2 Monthly Maximum Wind Speed Distribution

Identifies seasonal wind variations.

3 Rainfall vs Wind Speed Relationship

Highlights storm and cyclone activity correlation.

4 Seasonal Wind Speed Variation

Tracks wind behaviour across climatic seasons.

5 Wind Speed Variability Distribution

Measures turbulence and stability levels.

Dashboard 5 – Comprehensive Weather Insights

KPI Cards

Total Rainfall

Represents overall precipitation level.

Average Temperature

Displays general climate temperature trends.

Average Humidity

Shows atmospheric moisture concentration.

Average Wind Speed

Tracks wind behaviour across districts.

Extreme Weather Indicator

Highlights districts exposed to multiple extreme weather conditions.

Visual Insights

1 Multi-Parameter Weather Trend Analysis

Compares rainfall, temperature, humidity, and wind speed trends simultaneously.

2 Rainfall vs Temperature Relationship

Shows cooling effect of precipitation.

3 Seasonal Weather Comparison Matrix

Provides district-wise seasonal performance comparison.

4 Extreme Weather Event Distribution

Identifies high-risk districts exposed to environmental extremes.

5 Monthly Weather Metrics Distribution

Displays seasonal shifts across multiple weather parameters.

Tools & Technologies Used

- Power BI Desktop
- Power Query Editor
- DAX (Data Analysis Expressions)
- Data Modelling Techniques
- Time Intelligence Calculations

Key Challenges Faced

- **Data Quality Issues** : Multiple missing and inconsistent values required extensive cleaning and validation.
- **Seasonal Classification Complexity** : Mapping climate seasons required domain understanding and custom logic creation.
- **Handling Multi-Parameter Analysis** : Balancing rainfall, temperature, humidity, and wind simultaneously required optimized DAX measures.
- **Performance Optimization** : Large dataset size required measure optimization and model simplification.
- **Visualization Clarity** : Ensuring dashboards remain readable while handling multiple environmental metrics was challenging.

Business & Analytical Impact

This project demonstrates how weather data can be leveraged to:

- Identify climate change patterns
- Detect extreme weather events
- Support agriculture planning
- Improve disaster preparedness
- Assist environmental policy decisions

Project Summary

The Weather Analytics Dashboard successfully converts raw meteorological observations into actionable insights through advanced visualization and data modelling techniques. By combining rainfall, temperature, humidity, and wind speed analytics, this solution provides a comprehensive environmental monitoring system.

The project showcases strong capabilities in data cleaning, DAX calculations, visual storytelling, and dashboard design — closely reflecting real-world data analyst responsibilities.

Author

Developed as a real-world data analytics portfolio project demonstrating Power BI and environmental data analysis expertise.

Weather Analytics DAX Documentation

SECTION 1 – Calculated Columns

These columns are created inside the Weather table using **Modeling → New Column**

Time Intelligence Columns

Year =

`YEAR('Weather'[Date])`

Month Number =

`MONTH('Weather'[Date])`

Month Name =

`FORMAT('Weather'[Date], "MMMM")`

Quarter =

`"Q" & FORMAT('Weather'[Date], "Q")`

Season Classification Column

Season =

`SWITCH(
 TRUE(),
 MONTH('Weather'[Date]) IN {12,1,2}, "Winter",
 MONTH('Weather'[Date]) IN {3,4,5}, "Summer",
 MONTH('Weather'[Date]) IN {6,7,8,9}, "Monsoon",
 "Post-Monsoon"
)`

Temperature Derived Columns

Temperature Range =

`'Weather'[Max Temp (°C)] - 'Weather'[Min Temp (°C)]`

Humidity Derived Columns

Humidity Range =

`'Weather'[Max Humidity (%)] - 'Weather'[Min Humidity (%)]`

Wind Speed Derived Columns

Wind Speed Range =

'Weather'[Max Wind Speed (Kmph)] - 'Weather'[Min Wind Speed (Kmph)]

SECTION 2 – Rainfall Measures

Created using **Modeling → New Measure**

Rainfall Core Metrics

Total Rainfall =

SUM('Weather'[Rain (mm)])

Average Rainfall =

AVERAGE('Weather'[Rain (mm)])

Maximum Rainfall =

MAX('Weather'[Rain (mm)])

Annual Rainfall

Annual Rainfall =

CALCULATE(
 [Total Rainfall],
 ALLEXCEPT('Weather','Weather'[Year],'Weather'[District])
)

Rainfall Growth Percentage

Rainfall Growth % =

DIVIDE(
 [Total Rainfall] -
 CALCULATE([Total Rainfall], PREVIOUSYEAR('Weather'[Date])),
 CALCULATE([Total Rainfall], PREVIOUSYEAR('Weather'[Date]))
)

Monthly Rainfall Contribution %

Monthly Rainfall % =

DIVIDE(

[Total Rainfall],

CALCULATE([Total Rainfall], ALLEXCEPT('Weather','Weather'[Year],'Weather'[District]))

)

SECTION 3 – Temperature Measures

Average Temperature Metrics

Average Minimum Temperature =

AVERAGE('Weather'[Min Temp (°C)])

Average Maximum Temperature =

AVERAGE('Weather'[Max Temp (°C)])

Average Temperature Range =

AVERAGE('Weather'[Temperature Range])

Overall Average Temperature

Average Temperature =

AVERAGE(

('Weather'[Min Temp (°C)] + 'Weather'[Max Temp (°C)]) / 2

)

Extreme Temperature Measures

Highest Temperature Recorded =

MAX('Weather'[Max Temp (°C)])

Lowest Temperature Recorded =

MIN('Weather'[Min Temp (°C)])

SECTION 4 – Humidity Measures

Average Humidity Metrics

Average Minimum Humidity =

AVERAGE('Weather'[Min Humidity (%))

Average Maximum Humidity =

AVERAGE('Weather'[Max Humidity (%))

Average Humidity Range =

AVERAGE('Weather'[Humidity Range])

Extreme Humidity Metrics

Highest Humidity Recorded =

MAX('Weather'[Max Humidity (%))

Lowest Humidity Recorded =

MIN('Weather'[Min Humidity (%))

SECTION 5 – Wind Speed Measures

Average Wind Speed Metrics

Average Minimum Wind Speed =

AVERAGE('Weather'[Min Wind Speed (Kmph)])

Average Maximum Wind Speed =

AVERAGE('Weather'[Max Wind Speed (Kmph)])

Average Wind Speed Range =

AVERAGE('Weather'[Wind Speed Range])

Extreme Wind Speed Metrics

Highest Wind Speed Recorded =

MAX('Weather'[Max Wind Speed (Kmph)])

Lowest Wind Speed Recorded =

MIN('Weather'[Min Wind Speed (Kmph)])

SECTION 6 – Comprehensive Weather Measures

Average Weather Indicators

Average Humidity Overall =

AVERAGE('Weather'[Max Humidity (%))

Average Wind Speed Overall =

AVERAGE('Weather'[Max Wind Speed (Kmph)])

Extreme Weather Indicator

Extreme Weather Indicator =

IF(
 [Maximum Rainfall] > 100 ||
 [Highest Temperature Recorded] > 40 ||
 [Highest Wind Speed Recorded] > 50,
 "Extreme Event Detected",
 "Normal Conditions"
)

SECTION 7 – Trend Analysis Measures

Temperature Yearly Trend

Temperature Yearly Average =

CALCULATE(
 [Average Temperature],
 ALLEXCEPT('Weather','Weather'[Year],'Weather'[District])
)

Humidity Yearly Trend

Humidity Yearly Average =

CALCULATE(
 [Average Maximum Humidity],
 ALLEXCEPT('Weather','Weather'[Year],'Weather'[District])

)

Wind Yearly Trend

Wind Yearly Average =

CALCULATE(

[Average Maximum Wind Speed],

ALLEXCEPT('Weather','Weather'[Year],'Weather'[District])

)

SECTION 8 – Utility Sorting Column

Month Sort =

MONTH('Weather'[Date])

👉 Use this column to sort Month Name correctly.

SECTION 9 – Recommended Model Notes

- All calculations are created in the **Weather fact table**
 - Time intelligence requires **Date column as Date data type**
 - Month Name should be sorted by Month Sort column
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SECTION 10 – Documentation Summary

This DAX library supports:

- ✓ Rainfall Trend Analysis
- ✓ Temperature Behavioural Monitoring
- ✓ Humidity Pattern Evaluation
- ✓ Wind Speed Performance Tracking
- ✓ Multi-Parameter Weather Insights
- ✓ Extreme Weather Event Detection
- ✓ Seasonal Climate Pattern Analysis